

From the Binomial Theorem to Coq: An Educational Deciphering of Version 14 of the Conditional Model of Fermat's Last Theorem

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Abstract

This document presents an educational deciphering of version 14 of the conditional model around Fermat's Last Theorem. The text is written for a strong high school graduate of a mathematical lyceum (11th grade) and explains not the standard proof by Wiles, but a separate reconstructive scheme: the Fermat equation, parameterization via m, p , the odd binomial core, two real normalizations of this core, the condition of mutual scale symmetry $l^n = q^n$, the derivation of $n = 2^{n-1}$, and the final chain formally verified in Coq. The real, integer, and modular levels are considered separately so as not to mix equalities over $\mathbb{R}$ and $\mathbb{Z}$ with modular congruences. The main goal of the text is not to declare an unconditional new proof of Fermat's Last Theorem, but to clearly show the conditional logical structure: if two natural residual scales of the same odd binomial core satisfy mutual scale symmetry, then the exponent $n > 2$ is excluded.

Keywords: Fermat's Last Theorem; conditional mathematical model; binomial theorem; odd binomial core; real factorization; two core normalizations; mutual scale symmetry; $l^n = q^n$; $n = 2^{n-1}$; Coq; formal verification; modular caveat; integer specialization; educational exposition for a mathematical lyceum.

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A Preliminary Honest Remark on the Status of the Model

Fermat's Last Theorem has already been proven by modern mathematics in the works of Andrew Wiles and Richard Taylor. The modern proof uses deep methods of modular forms and elliptic curves. This educational text does not replace that proof and is not an unconditional new proof of the theorem.

A different task is considered here: to carefully present a conditional reconstructive model, in which the odd binomial core is highlighted and two natural real normalizations of this core are compared. Therefore, the key difference must be stated immediately:

The unconditional proof asserts FLT without additional hypotheses. **The conditional model** proves an implication: if the explicitly specified condition of mutual scale symmetry $l^n = q^n$ is satisfied, then the exponent $n > 2$ is impossible.

There are two parts to this model:

1. **Algebraic-reconstructive part:** from the symmetric notation of the difference of powers, the odd binomial core $S_n(m, p)$ and its two normalizations are extracted.
2. **Formally verified part:** after accepting the two factorizations and the condition $l^n = q^n$, Coq verifies the derivation of $n = 2^{n-1}$, and then $n \in \{1, 2\}$.

Such a separation is important. Computer verification does not replace the mathematical justification for the central condition $l^n = q^n$, but it guarantees that further logical transitions within the chosen conditional scheme are executed rigorously.

What is Changed in Version 14

The current version is built around a new central scheme:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n, \quad l^n = q^n \implies n = 2^{n-1} \implies n \in \{1, 2\}.$$

The main changes are as follows.

1. The main object becomes the odd binomial core $S_n(m, p)$.
2. Instead of a single scale parameter, two residual scales are used: l and q .
3. The first normalization extracts the linear factor n :

$$S_n(m, p) = n l^n.$$

4. The second normalization extracts the mass of odd binomial coefficients 2^{n-1} :

$$S_n(m, p) = 2^{n-1} q^n.$$

5. The central conditional point is the mutual scale symmetry:

$$l^n = q^n.$$

6. The modular level is presented separately. The real scale q is not a modulus; the notation $\text{mod } q$ is used for the modulus.

7. The Coq description is provided in terms of the current file `GlobalNormalization.v`: `real_core_factorization`, `coefficient_mass_factorization`, `mutual_scale_symmetry`, `MutualScaleData`, and related lemmas.

For a graduate of a mathematical lyceum, this text is useful as an example of mathematical culture: a strong idea must be written in such a way that definitions, the domain of applicability, conditional points, model boundaries, and machine-verifiable consequences are all visible.

1 Historical Statement of the Problem

Fermat's Last Theorem states that the equation

$$x^n + y^n = z^n \tag{1}$$

has no solutions in positive integers x, y, z for integer $n > 2$.

For $n = 1$, solutions are obvious: for example, $1 + 2 = 3$. For $n = 2$, Pythagorean triples exist:

$$3^2 + 4^2 = 5^2.$$

But starting with cubes, the situation changes: one cannot represent a cube as the sum of two cubes, a fourth power as the sum of two fourth powers, and so on.

Fermat formulated this statement in the margins of his copy of Diophantus's *Arithmetica* and wrote that he had found a "truly marvelous demonstration," but the margins of the book were too narrow to contain it. From a historical point of view, it is natural to ask: could Fermat have had an elementary idea that he did not write down in modern form?

This document does not claim that the historical proof of Fermat has been found. It offers an educational reconstruction of a conditional scheme showing how a rigorous implication can emerge from the binomial theorem, the symmetry of two terms, and the comparison of two real scales.

2 How a Mathematical Lyceum Graduate Should Read This Document

To understand the text, it is sufficient to confidently command the following topics:

- powers, roots, and natural numbers;
- the binomial theorem;
- elementary divisibility;
- comparing function growth rates;
- a basic understanding of what formal proof verification is.

The main methodological principle: we will distinguish between three things.

1. **Algebraic identity**: an equality obtained by expanding brackets or substituting variables.
2. **Model condition**: an additional statement that needs to be accepted or proven separately.
3. **Formal deduction**: a consequence that Coq verifies after all prerequisites are explicitly written down.

This distinction protects against a typical mistake in amateur proofs: a beautiful observation cannot be imperceptibly turned into a proven theorem. In this version, the central conditional point is named explicitly: it is the condition $l^n = q^n$.

3 Map of Correspondence Between the Article, Educational Text, and Coq

So as not to get lost in the notation, it is useful to keep the following map in front of your eyes.

Level	Mathematical Meaning	Name in Coq
Real parameterization	If $z = m^n + p^n$, $x = m^n - p^n$, then $z + x = 2m^n$, $z - x = 2p^n$ over $\mathbb{R}$.	sum_diff_from_parameters_R
Integer specialization	The same formula over $\mathbb{Z}$; with integer m, p , parity conditions arise.	sum_diff_from_parameters_Z; parity_condition_Z
Checking the impossibility of integer parameters	If $z + x$ or $z - x$ has the wrong parity, there are no integer m, p in such a form.	no_parameters_if_parity_violation; no_parameters_if_odd; no_parameters_for_example
First core normalization	The odd core is written as $\text{core} = n l^n$ over $\mathbb{R}$.	real_core_factorization
Second core normalization	The same core is written as $\text{core} = 2^{n-1} q^n$ over $\mathbb{R}$.	coefficient_mass_factorization
Comparison of two normalizations	From two expressions of the same core follows $l^n = q^n \frac{2^{n-1}}{n}.$	two_scale_factorizations_relation
Mutual scale symmetry	The central condition $l^n = q^n$.	mutual_scale_symmetry
Transition to the exponential-linear equation	Two factorizations, $l^n = q^n$ and $q > 0$ entail $n = 2^{n-1}$.	mutual_scale_symmetry_forces_shift
Packaging the model data	One structure stores n , the core, l , q , positivity, and both factorizations.	MutualScaleData
Final growth	From $n = 2^{n-1}$ follows $n = 1$ or $n = 2$.	binary_scaling_roots_only_one_two
Exclusion of large exponents	From the model data and $n > 2$, a contradiction follows.	mutual_scale_data_excludes_high_exponent
Modular caveat	An integer equality entails a congruence modulo, but the reverse is false.	ModularRemark; modular_congruence_not_integer_equality

The educational text explains the idea in human language. The article fixes the mathematical formulation. The Coq file verifies only those implications that are explicitly written in the code.

4 Parameterization via m and p

Suppose hypothetically there exists a positive integer solution to the Fermat equation for some $n > 2$:

$$x^n + y^n = z^n.$$

Let us rewrite it as

$$z^n - x^n = y^n. \quad (2)$$

Now we introduce the parameters m and p using the system

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \quad (3)$$

Let us add and subtract these equalities:

$$2m^n = z + x, \quad 2p^n = z - x.$$

From here, formally

$$m = \sqrt[n]{\frac{z+x}{2}}, \quad p = \sqrt[n]{\frac{z-x}{2}}. \quad (4)$$

Remark 4.1. If m and p are allowed as real numbers, such a reconstruction is an algebraic notation after choosing a suitable branch of the root. However, if one additionally requires $m, p \in \mathbb{Z}$, conditions arise: $z+x$ and $z-x$ must be even, and the numbers $(z+x)/2$ and $(z-x)/2$ must be perfect n -th powers in $\mathbb{Z}$.

Substituting (3) into (2), we get

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n. \quad (5)$$

It is here that the binomial part begins.

5 Three Levels of Numbers: $\mathbb{R}$, $\mathbb{Z}$, and the Modular Level

The same expression can live in different mathematical worlds. In this model, it is important to distinguish three levels.

5.1 Real level

Here $m, p, l, q \in \mathbb{R}$. One can extract roots, introduce real scales, and write down factorizations of the form

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

At this level, the notation implies equality of real quantities.

5.2 Integer level

Here, conditions like $x, y, z \in \mathbb{N}$ and, if required, $m, p \in \mathbb{Z}$ are additionally discussed. For example, from

$$z = m^n + p^n, \quad x = m^n - p^n$$

for $m, p \in \mathbb{Z}$ it follows that

$$z + x = 2m^n, \quad z - x = 2p^n.$$

Consequently, $z + x$ and $z - x$ must be even. If this is not the case, there can be no integer parameters m, p in such a form.

Example 5.1. Let $n = 3$, $z = 2$, $x = 1$. Then

$$\frac{z + x}{2} = \frac{3}{2}, \quad \frac{z - x}{2} = \frac{1}{2}.$$

Real roots exist, but there are no integer m, p satisfying

$$2 = m^3 + p^3, \quad 1 = m^3 - p^3.$$

It is exactly this type of difference that the lemma `no_parameters_for_example` captures.

5.3 Modular level

At the modular level, remainders are considered. To avoid confusing the modulus with the real scale q , we will denote the modulus by `modq`:

$$x^n + y^n \equiv z^n \pmod{\text{modq}}.$$

Such a congruence can be true even when the equality $x^n + y^n = z^n$ in integers is false. Therefore, modular examples in themselves are not counterexamples to FLT.

6 The Binomial Theorem and the Odd Binomial Core

Recall the binomial theorem:

$$(A + B)^n = \sum_{r=0}^n \binom{n}{r} A^{n-r} B^r. \quad (6)$$

For the difference we have

$$(A - B)^n = \sum_{r=0}^n (-1)^r \binom{n}{r} A^{n-r} B^r. \quad (7)$$

Subtracting (7) from (6), we get

$$(A + B)^n - (A - B)^n = \sum_{r=0}^n (1 - (-1)^r) \binom{n}{r} A^{n-r} B^r \quad (8)$$

$$= 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} A^{n-r} B^r. \quad (9)$$

In our case

$$A = m^n, \quad B = p^n.$$

Therefore,

$$y^n = 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r. \quad (10)$$

Definition 6.1 (Odd binomial core). The odd binomial core will refer to the sum

$$S_n(m, p) = \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r. \quad (11)$$

Then

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p), \quad (12)$$

and, consequently,

$$y^n = 2S_n(m, p).$$

Intuitively, $S_n(m, p)$ is the part of the binomial expansion that survives after subtracting two symmetric powers. Even terms are destroyed, and odd ones are doubled.

7 First Real Normalization: $S_n(m, p) = n l^n$

For every odd index r , the binomial coefficient can be written as

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}. \quad (13)$$

This allows us to extract the common factor n from the odd core in real-rational algebra.

Therefore, the first normalization is introduced:

$$S_n(m, p) = n l^n. \quad (14)$$

Here $l > 0$ is the real residual scale. It shows what part of the core remains after extracting the linear factor n .

If $n > 0$, then formally

$$l^n = \frac{S_n(m, p)}{n}. \quad (15)$$

When the right-hand expression is positive, one can take the positive real branch of the root.

From $y^n = 2S_n(m, p)$ and (14) we obtain

$$y^n = 2n l^n. \quad (16)$$

In Coq, this corresponds to the definition `real_core_factorization`. It means:

$$0 < n, \quad \text{core} = n l^n$$

over the real numbers.

8 Why This is Not Integer Divisibility

The notation

$$S_n(m, p) = n l^n$$

over $\mathbb{R}$ does not automatically imply

$$n \mid S_n(m, p)$$

in $\mathbb{Z}$.

This is a fundamentally important distinction. The first statement is a real definition of the scale l . The second statement is integer divisibility, which requires a separate proof and may not be true at all.

Example 8.1 (Check for $n = 6$). For $n = 6$, $m = 1$, $p = 1$, the odd sum of binomial coefficients is equal to

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

The number 32 is not divisible by 6 in $\mathbb{Z}$. But over $\mathbb{R}$ one can write

$$32 = 6 \cdot \frac{16}{3} = 6l^6,$$

if we set

$$l^6 = \frac{16}{3}.$$

This is not a problem for the real model, because it uses equality over $\mathbb{R}$, not a statement about divisibility in the ring of integers.

Therefore, the correct formulation is as follows:

$$l^n = \frac{S_n(m, p)}{n} \quad \text{as a real definition,}$$

but not necessarily

$$n \mid S_n(m, p) \quad \text{as an integer statement.}$$

9 Second Real Normalization: $S_n(m, p) = 2^{n-1}q^n$

Now let's look not at the factor n , but at the total mass of the odd binomial coefficients.

From the binomial identity, it is known:

$$\sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} = 2^{n-1}. \quad (17)$$

This means: the sum of odd coefficients in a row of Pascal's triangle is equal to half of the total sum of coefficients.

Therefore, we can introduce a second positive real residual scale $q > 0$:

$$S_n(m, p) = 2^{n-1}q^n. \quad (18)$$

Equivalently,

$$q^n = \frac{S_n(m, p)}{2^{n-1}}. \quad (19)$$

Here q is not a modulus and not a remainder modulo. It is a real scale parameter that remains after extracting the mass of the odd binomial coefficients 2^{n-1} .

From $y^n = 2S_n(m, p)$ and (18) we obtain

$$y^n = 2^n q^n. \quad (20)$$

In Coq, this normalization corresponds to the definition `coefficient_mass_factorization`. Its meaning is:

$$0 < n, \quad \text{core} = 2^{n-1} q^n$$

over the real numbers.

10 Comparison of the Two Normalizations

Now the same core $S_n(m, p)$ is written in two ways:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n. \quad (21)$$

Equating the right-hand sides, we get

$$n l^n = 2^{n-1} q^n.$$

Since $n > 0$, we can divide by n :

$$l^n = q^n \frac{2^{n-1}}{n}. \quad (22)$$

This is not a hypothesis, but a direct consequence of two real factorizations of the exact same core. This is exactly what the Coq lemma `two_scale_factorizations_relation` verifies.

Substantively, formula (22) says: two normalizations of a single core generally yield different residual scales. Their ratio is governed by the factor

$$\frac{2^{n-1}}{n}.$$

11 The Condition of Mutual Scale Symmetry $l^n = q^n$

The central condition of the current model is formulated as follows:

$$l^n = q^n. \quad (23)$$

Let us call it the **condition of mutual scale symmetry**. It means that two natural residual scales of the same symmetric two-power expansion coincide at the level of n -th powers.

It is important to understand the status of this condition. It is not a consequence solely of expanding the binomial theorem. It is precisely this that constitutes the substantive mathematical point of the model. It must either be proven from the internal binary symmetry of a hypothetical solution, or left as an explicitly stated structural hypothesis.

In Coq, this condition is written as `mutual_scale_symmetry n l q`, i.e., literally:

$$l^n = q^n.$$

12 Derivation of $n = 2^{n-1}$

From comparing the two normalizations we already obtained

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If the condition of mutual scale symmetry is additionally satisfied

$$l^n = q^n,$$

we obtain

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, then $q^n > 0$. Hence, it is possible to cancel q^n :

$$1 = \frac{2^{n-1}}{n}.$$

Consequently,

$$n = 2^{n-1}. \tag{24}$$

It is exactly this transition that the Coq lemma `mutual_scale_symmetry_forces_shift` verifies. It asserts: if the two factorizations hold

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

if $l^n = q^n$ holds, and if $q > 0$, then

$$n = 2^{n-1}.$$

13 Why $n = 2^{n-1}$ Yields Only $n = 1, 2$

Consider the equation

$$n = 2^{n-1}.$$

For $n = 1$ we get

$$1 = 2^0 = 1.$$

For $n = 2$ we get

$$2 = 2^1 = 2.$$

That is, $n = 1$ and $n = 2$ are indeed solutions.

Let us show that there are no solutions for $n > 2$. Equivalently, it can be written as

$$2^n = 2n.$$

But for all integers $n \geq 3$ we have

$$2^n > 2n. \tag{25}$$

For example:

$$2^3 = 8 > 6, \quad 2^4 = 16 > 8, \quad 2^5 = 32 > 10.$$

Further on, the gap only grows, because the exponential function doubles at each step, while the linear function adds only a constant amount.

In the Coq language, this part is split into several elementary lemmas:

- `pow2_gt_linear_shift;`
- `pow2_gt_linear;`

- pow_eq_linear_positive;
- two_pow_eq_two_mul_iff_shift;
- binary_scaling_roots_only_one_two.

The final formal thought is simple:

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

14 Conditional Theorem of the Model

Theorem 14.1 (Conditionally under mutual scale symmetry). *Suppose a hypothetical positive integer solution to the Fermat equation*

$$x^n + y^n = z^n, \quad n > 2,$$

admits a real symmetric parameterization

$$z = m^n + p^n, \quad x = m^n - p^n.$$

Suppose the odd binomial core $S_n(m, p)$ admits two real normalizations

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

where $l > 0, q > 0$. If the condition of mutual scale symmetry is additionally satisfied

$$l^n = q^n,$$

then the exponent $n > 2$ is impossible.

Proof. From the two normalizations follows

$$l^n = q^n \frac{2^{n-1}}{n}.$$

By the condition $l^n = q^n$, therefore

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, then $q^n > 0$, and cancellation by q^n is legitimate. We obtain

$$n = 2^{n-1}.$$

But this equation in positive integers only has solutions $n = 1$ and $n = 2$. Consequently, $n > 2$ contradicts the specified premises. $\square$

This theorem is conditional. The main mathematical question is whether $l^n = q^n$ can be derived from the very internal structure of a hypothetical solution to the Fermat equation.

15 The Modular Caveat

In discussions of FLT, examples of congruences of the form

$$a^n + b^n \equiv c^n \pmod{\text{mod}q}$$

are often cited. Such examples can be correct, but they relate to a different level of the problem.

Congruence modulo means that there exists an integer t such that

$$c^n = a^n + b^n + \text{mod}q \cdot t.$$

Equality in integers is only a special case where $t = 0$.

Example 15.1. We have

$$1^3 + 1^3 = 2, \quad 3^3 = 27.$$

In integers $2 \neq 27$. But modulo 5

$$27 = 2 + 5 \cdot 5,$$

hence

$$3^3 \equiv 1^3 + 1^3 \pmod{5}.$$

This is a true modular congruence, but not a true integer equality.

Therefore, the existence of solutions to congruences

$$x^n + y^n \equiv z^n \pmod{\text{mod}q}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not refute the real parameterization.

If specifically transitioning to the modular level, parameters must be set separately:

$$m^n \equiv A \pmod{\text{mod}q}, \quad p^n \equiv B \pmod{\text{mod}q},$$

and only then write

$$z \equiv A + B \pmod{\text{mod}q}, \quad x \equiv A - B \pmod{\text{mod}q}.$$

16 How the Modular Caveat is Reflected in Coq

In Coq, there is a section `ModularRemark`. In it, the relation of modular congruency is defined as

$$\text{modular_congruent mod}q \ a \ b.$$

Mathematically, this means:

$$\exists t \in \mathbb{Z} : \quad a = b + \text{mod}q \cdot t.$$

The lemma `integer_equality_implies_modular_power_congruence` verifies the direction:

$$\text{integer equality} \implies \text{modular congruence}.$$

If

$$x^n + y^n = z^n,$$

then one can take $t = 0$.

The lemma `integer_equality_is_zero_modular_witness` explicitly fixes that a true equality corresponds to a zero modular witness $t = 0$.

The lemma `modular_congruence_not_integer_equality` verifies the example:

$$3^3 \equiv 1^3 + 1^3 \pmod{5},$$

but

$$3^3 \not\equiv 1^3 + 1^3$$

in integers.

The structure `ModularParameterResidues` formalizes the idea that at the modular level, one must separately set the residues of m^n and p^n , and then separately set the residues of z and x .

17 Connection with the Main Coq Fragments

This section lists the main Coq objects and their meanings.

`sum_diff_from_parameters_R`

Verifies $z + x = 2m^n$ and $z - x = 2p^n$ over $\mathbb{R}$.

`sum_diff_from_parameters_Z`

The same over $\mathbb{Z}$.

`parity_condition_Z`

If $m, p \in \mathbb{Z}$, then $z + x$ and $z - x$ must be even.

`no_parameters_if_parity_violation`

If the parity of $z + x$ or $z - x$ is violated, there are no integer parameters m, p in such a form.

`no_parameters_if_odd`

The same idea written through the oddness of the respective sums and differences.

`no_parameters_for_example`

A formal example $z = 2$, $x = 1$, $n = 3$: real reconstruction is possible, but integer reconstruction is impossible.

`real_core_factorization`

The first normalization core $= n l^n$.

`coefficient_mass_factorization`

The second normalization core $= 2^{n-1} q^n$.

`mutual_scale_symmetry`

The condition of mutual scale symmetry $l^n = q^n$.

`two_scale_factorizations_relation`

From two factorizations of one core, derives the exact ratio $l^n = q^n \frac{2^{n-1}}{n}$.

`mutual_scale_symmetry_forces_shift`

From two factorizations, the condition $l^n = q^n$, and positivity $q > 0$, derives $n = 2^{n-1}$.

`MutualScaleData`

Packages n , core, l , q , positivity, and the model conditions.

mutual_scale_data_forces_shift

From MutualScaleData derives $n = 2^{n-1}$.

binary_scaling_roots_only_one_two

From $n = 2^{n-1}$ derives $n = 1$ or $n = 2$.

mutual_scale_data_forces_small_exponent

From the model data derives $n = 1$ or $n = 2$.

mutual_scale_data_excludes_high_exponent

From the model data and $n > 2$ derives a contradiction.

18 What Exactly Coq Verifies

Coq does not verify the entire historical hypothesis nor the unconditional FLT. It verifies the formal conditional chain:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n, \quad l^n = q^n, \quad q > 0$$

entail

$$n = 2^{n-1}.$$

Then Coq verifies the elementary arithmetic part:

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

Consequently,

$$n > 2$$

is impossible under these premises.

In other words, Coq confirms the absence of logical errors in the final conditional deduction. But the actual substantive reason why $l^n = q^n$ should hold for a hypothetical solution needs to be discussed separately.

19 What Coq Does Not Prove

To be honest, it must be explicitly stated: Coq does not automatically prove that from every hypothetical solution of the equation

$$x^n + y^n = z^n$$

it follows that

$$l^n = q^n.$$

This condition is the central structural premise of the current model. Coq verifies something else: if this condition is accepted along with the two factorizations of the core, then the further deduction is rigorous.

Therefore, the weak point of the conditional scheme lies neither in the arithmetic of the equation $n = 2^{n-1}$ nor in the final chain verified by Coq. The decisive point remains the mathematical status of the condition of mutual scale symmetry.

20 Why Compilation Time is Important, But Does Not Replace Mathematics

If a Coq file compiles in a few seconds, this means that the formally written lemmas and theorems have been verified by a machine. This is a good technical sign: there are no missing proofs, illegal transitions, or unclosed goals in the code.

However, compilation time is not a proof of the initial mathematical hypothesis. The machine only verifies what is written in the formal language. If the central condition $l^n = q^n$ is introduced as a premise, Coq honestly verifies the consequences of this premise, but does not derive it out of thin air.

Therefore, the correct formula is this:

Fast Coq compilation confirms the correctness of the formal deduction within the specified premises, but does not turn a conditional premise into an unconditional theorem.

21 How to Correctly Formulate a Message for a Forum

For a mathematical forum, it is better to use calm and precise phrasing:

I propose to discuss a conditional reconstructive model around FLT. In it, the odd binomial core $S_n(m, p)$ is written in two ways: $S_n(m, p) = n l^n$ and $S_n(m, p) = 2^{n-1} q^n$. From comparing these normalizations it follows that $l^n = q^{n \frac{2^n - 1}{n}}$. If the condition of mutual scale symmetry $l^n = q^n$ is additionally met, then we get $n = 2^{n-1}$, whence $n \in \{1, 2\}$. Coq verifies the final formal chain, but the main question remains mathematical: how well-founded is the condition $l^n = q^n$ for a hypothetical solution to the Fermat equation?

Such a style is honest and constructive. It does not require the reader to immediately accept the whole model, but invites them to verify a specific central point.

22 Typical Questions and Answers

22.1 Does this educational file prove FLT unconditionally?

No. This is an educational explanation of a conditional model. Unconditional FLT was proven by Wiles and Taylor. A different conditional scheme is considered here.

22.2 What is the main conditional point?

The main conditional point is the mutual scale symmetry:

$$l^n = q^n.$$

It is exactly this that needs to be justified separately.

22.3 What exactly does Coq verify?

Coq verifies the formal chain:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n, \quad l^n = q^n, \quad q > 0 \implies n = 2^{n-1} \implies n \in \{1, 2\}.$$

22.4 Why is q not a modulus?

In this model, q is the real residual scale of the second normalization. The modulus in modular arithmetic is denoted by $\text{mod } q$.

22.5 Why is real factorization not divisibility?

Because

$$S_n(m, p) = n l^n$$

over $\mathbb{R}$ means only real equality. It does not automatically mean

$$n \mid S_n(m, p)$$

in $\mathbb{Z}$.

22.6 Why can't the condition $l^n = q^n$ simply be declared obvious?

Because it is exactly the substantive mathematical point of the model. The two normalizations of one core give the exact ratio

$$l^n = q^n \frac{2^{n-1}}{n}.$$

To obtain $n = 2^{n-1}$, one needs to additionally require $l^n = q^n$. This should not be a hidden transition.

22.7 Why is the quadratic case special?

Because the equation

$$n = 2^{n-1}$$

has positive solutions only for $n = 1$ and $n = 2$. Therefore, within the framework of this conditional scheme, the linear and quadratic cases remain permissible, while exponents higher than a square are excluded.

23 Educational Exercises

1. Expand $(A + B)^5 - (A - B)^5$ and verify that only odd powers of B remain.
2. For $n = 4$, write out the odd core $S_4(m, p)$ explicitly.
3. Prove the identity

$$\sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} = 2^{n-1}$$

using the equalities $(1 + 1)^n$ and $(1 - 1)^n$.

4. For $n = 6$, $m = 1$, $p = 1$, verify that the odd sum of coefficients is 32, and explain why this is not divisible by 6 in $\mathbb{Z}$.
5. Show by induction that for all $n \geq 3$, it is true that $2^n > 2n$.
6. Explain in your own words how an equality over $\mathbb{Z}$ differs from a modular congruence.
7. Formulate in your own words what exactly the condition $l^n = q^n$ means.

24 Final Scheme of the Entire Model

In a compressed form, the entire conditional scheme looks like this:

$$\begin{aligned}
 & x^n + y^n = z^n \\
 & \Downarrow \\
 & y^n = (m^n + p^n)^n - (m^n - p^n)^n \\
 & \Downarrow \\
 & (m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p) \\
 & \Downarrow \\
 & S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n \\
 & \Downarrow \\
 & l^n = q^n \frac{2^{n-1}}{n} \\
 & \Downarrow \quad \text{under the condition } l^n = q^n \\
 & n = 2^{n-1} \\
 & \Downarrow \\
 & n \in \{1, 2\} \\
 & \Downarrow \\
 & \text{for } n > 2 \text{ a contradiction arises.}
 \end{aligned}$$

25 Conclusion

Version 14 of the educational file explains the scheme of mutual scale symmetry of two residual scales l and q . The main logic consists in comparing two natural real normalizations of the very same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

From them follows the exact ratio

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If the condition

$$l^n = q^n,$$

is additionally satisfied, then we get

$$n = 2^{n-1},$$

and this is only possible for $n = 1$ and $n = 2$.

Coq verifies the final formal chain, but does not replace the proof of the central condition $l^n = q^n$. Therefore, the correct status of the model is a conditional reconstruction: if mutual scale symmetry is satisfied, the exponent $n > 2$ is impossible.

26 Mini-Glossary

Term	Meaning
Fermat's Last Theorem	The statement that $x^n + y^n = z^n$ has no positive integer solutions for $n > 2$.
Binomial Theorem	The formula for expanding $(A + B)^n$ via binomial coefficients.
Odd binomial core	The sum of odd terms remaining in $(m^n + p^n)^n - (m^n - p^n)^n$ after canceling the even terms.
Real factorization	A notation of the form $S_n(m, p) = n l^n$ over $\mathbb{R}$; not the same as divisibility in $\mathbb{Z}$.
Residual scale l	The real scale after extracting the linear factor n .
Residual scale q	The real scale after extracting the mass of odd coefficients 2^{n-1} .
Modulus mod q	The number modulo which remainders are considered; it is not the scale q .
Mutual scale symmetry	The condition $l^n = q^n$.
Coq	An interactive theorem proving system that verifies formal deductions.

27 List of Sources and References

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